

1

GENERAL PRINCIPLES AND CHARACTERISTICS OF OPTICAL MAGNETOMETERS

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1.1 Introduction

Optical magnetometry encompasses the wide range of experimental techniques in which light is used to measure the response of atomic angular momentum to magnetic fields. Atomic magnetic moments $\boldsymbol{\mu}$ arise due to the magnetic moments associated with the intrinsic spins of constituent electrons and nuclei as well as electronic orbital motion. In the presence of a magnetic field $\mathbf{B}$, a torque

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$$

acts on the atoms. For sufficiently small magnetic fields the atomic magnetic moment $\boldsymbol{\mu}$ can be assumed to be independent of $\mathbf{B}$, in which case $\boldsymbol{\tau}$ causes the component of angular momentum transverse to $\mathbf{B}$ to precess about $\mathbf{B}$ at the Larmor frequency $\Omega_L = \gamma B$, where γ is the gyromagnetic ratio of the atomic species. When light propagates through and interacts with the atomic medium, angular momentum is exchanged between the atoms and the light field. Thus the angular momentum state of the atoms affects the angular momentum state of the light. In this way, precession of atomic angular momentum can be observed via the induced changes in the polarization and intensity of light interacting with the atoms, allowing optical measurement of Larmor precession.

The exchange of angular momentum between the atomic medium and light field that enables optical detection of Larmor precession can similarly be used to polarize the atomic medium through the process of *optical pumping* (see the comprehensive reviews [1–3]). Optical pumping refers to the process where photons absorbed by atoms transfer their angular momentum to the atoms, thereby creating polarization in both ground and excited states. Ground-state spin polarization is achieved through both depopulation of selected Zeeman sublevels and repopulation of Zeeman sublevels by spontaneous transitions from the excited states polarized by the incident light. In most optical magnetometers, the signals used to detect Larmor precession scale with the ground-state spin polarization, and consequently, optical pumping dramatically enhances magnetometer sensitivity.

Optical magnetometers, presently unmatched in both absolute accuracy and mag-

netometric sensitivity [4, 5], are universally based on these aforementioned principles: (1) optical pumping of atomic spin polarization, (2) time-evolution of atomic spin polarization due to the torque exerted on atomic magnetic moments by the magnetic field, and (3) optical detection of the evolved atomic spin polarization state through the effect of the polarized atoms on light propagating through the atomic medium. Within this basic framework, there is a remarkable diversity of experimental techniques and physical effects, many of which are explored throughout this book, the cited references, and online supporting material (available at www.cambridge.org/9781107010352), and undoubtedly still others that remain to be discovered.

1.1.1 Fundamental sensitivity limits

The fundamental quantum mechanical uncertainty in the measurement of atomic spin projection constrains the potential sensitivity of optical magnetometers. The *spin-projection noise limited* (or *atomic shot noise limited*) sensitivity δB_{SNL} of a polarized atomic sample to magnetic fields¹ is determined by the total number of atoms N and the spin-relaxation rate Γ_{rel} for measurement times $\tau \gg \Gamma_{\text{rel}}^{-1}$ [6]:

$$\delta B_{\text{SNL}} \approx \frac{1}{\gamma} \sqrt{\frac{\Gamma_{\text{rel}}}{N\tau}}. \quad (1.1)$$

Equation (1.1) can be understood by noting that a measurement of a single atomic spin for a time $1/\Gamma_{\text{rel}}$ determines the Larmor precession angle with an uncertainty on the order of one radian. If the measurement is performed with N atoms, the uncertainty is reduced by $\sqrt{N}$, and if the measurement is repeated multiple times, the uncertainty is reduced by the square root of the number of measurements, which most efficiently is $\sqrt{\Gamma_{\text{rel}}\tau}$.

As can be seen from Eq. (1.1), to achieve the highest possible precision in magnetometric measurements, it is advantageous to have the longest possible relaxation time for the atomic polarization, i.e., the smallest possible Γ_{rel} , as well as the largest possible N . Therefore optical magnetometers typically are based on measurements of long-lived ground-state spin polarization [an exception is the ^4He magnetometer (Chapter 10), in which polarization of the metastable 2^3S_1 state is used]. A variety of techniques can be employed to minimize Γ_{rel} : antirelaxation coating the walls of the vapor cell containing the atoms to reduce spin-depolarizing wall collisions [7–10] (see Chapter 11), filling the vapor cell with buffer gas to slow diffusion of atoms to the cell walls (see, for example, Refs. [11–15]), and even atom trapping and cooling [16, 17] (Chapter 9). Also of note are optical magnetometers using particular condensed matter systems, such as nitrogen-vacancy centers in diamonds (Chapter

¹ Here we ignore factors of order unity that depend on the details of the atomic system and optical magnetometer scheme, such as the total atomic angular momentum and relative contributions of different Zeeman sublevels.

8) and alkali atoms trapped in condensed (superfluid or solid) helium [18–20], that have small Γ_{rel} .

There is also a contribution to optical-magnetometer noise from the quantum uncertainty of measurements of light properties (photon shot noise). Optical detection of atomic spin precession is usually performed by measuring either the intensity or polarization of light transmitted through the atomic sample. There are certain intrinsic advantages to measuring the light polarization, in particular, reduced sensitivity to noise due to laser-intensity fluctuations. If, for example, atomic spin precession is detected by measuring optical rotation of the plane of transmitted light polarization [21], the photon-shot-noise-limited sensitivity to the optical rotation angle φ is

$$\delta\varphi \approx \frac{1}{2} \sqrt{\frac{1}{\Phi\tau}} , \quad (1.2)$$

where Φ is the probed photon flux (photons/s) detected after the atomic sample and $\delta\varphi$ is measured in $\text{rad}/\sqrt{\text{Hz}}$. It should also be noted that light-atom coupling via AC Stark shifts can generate additional noise (see Ref. [22] and Sec. 1.4.3). In optimal operation, the contribution of photon shot noise to overall magnetometric noise does not exceed the contribution from atomic spin-projection noise [6, 23]. Upon optimization of the atomic density n for a given volume V of the sample, where $N = nV$, for an atomic-vapor-based optical magnetometer the dominant spin-relaxation mechanism becomes either spin-exchange or spin-destruction collisions (depending on the details of the magnetometry scheme), in which case $\Gamma_{\text{rel}} = \xi n$, and the optimum magnetometric sensitivity becomes

$$\delta B_{\text{opt}} \approx \frac{1}{\gamma} \sqrt{\frac{\xi}{V\tau}} . \quad (1.3)$$

The relaxation constant ξ ranges between $\sim 10^{-9} \text{ cm}^3/\text{s}$ and $\sim 10^{-13} \text{ cm}^3/\text{s}$ for alkali atoms, depending on the details of the collisions [1]. Thus for optical magnetometers using alkali vapors, the optimal magnetometric sensitivity for a $V = 1 \text{ cm}^3$ magnetic sensor ranges between 10^{-11} to $10^{-13} \text{ G}/\sqrt{\text{Hz}}$ (1 to 0.01 fT/ $\sqrt{\text{Hz}}$). Issues related to noise in optical magnetometers, including exploration of squeezed states and quantum nondemolition (QND) measurements, are addressed in detail in Chapters 2 and 3.

1.1.2 Zeeman shifts and atomic spin precession

The language of atomic spectroscopy provides a complementary description of optical magnetometry: the light field propagating through the atomic medium measures the Zeeman shifts of atomic states. The Hamiltonian describing the Zeeman shift is

$$H_Z = -\boldsymbol{\mu} \cdot \mathbf{B} . \quad (1.4)$$

For an elementary particle such as the electron the magnetic moment is directly proportional to the intrinsic spin $\mathbf{S}$, $\boldsymbol{\mu} = g_e \mu_B \mathbf{S}$, where g_e is the electron's Landé g-factor and μ_B is the Bohr magneton,

$$\mu_B = \frac{e\hbar}{2mc} = \frac{\alpha}{2} e a_0 \approx (2\pi\hbar) \times 1.4 \text{ MHz/G} , \quad (1.5)$$

where e is the absolute value of the electron charge, $\hbar$ is Planck's constant, m is the electron mass, c is the speed of light, $\alpha = e^2/(\hbar c)$ is the fine structure constant, and $a_0 = \hbar^2/(me^2)$ is the Bohr radius. The Landé g-factor for the electron, $g_e \approx 2.00232$, can be determined from extraordinarily precise experimental measurements [24–26] and from the theory of quantum electrodynamics [27], and thus, measurements of the Zeeman shifts of electron spin states constitute absolute measurements of the magnetic field.

For an atom the situation becomes more complicated. For sufficiently small magnitudes of $\mathbf{B}$, the atomic magnetic moment $\boldsymbol{\mu}$ is determined by internal atomic interactions and is approximately independent of $\mathbf{B}$,

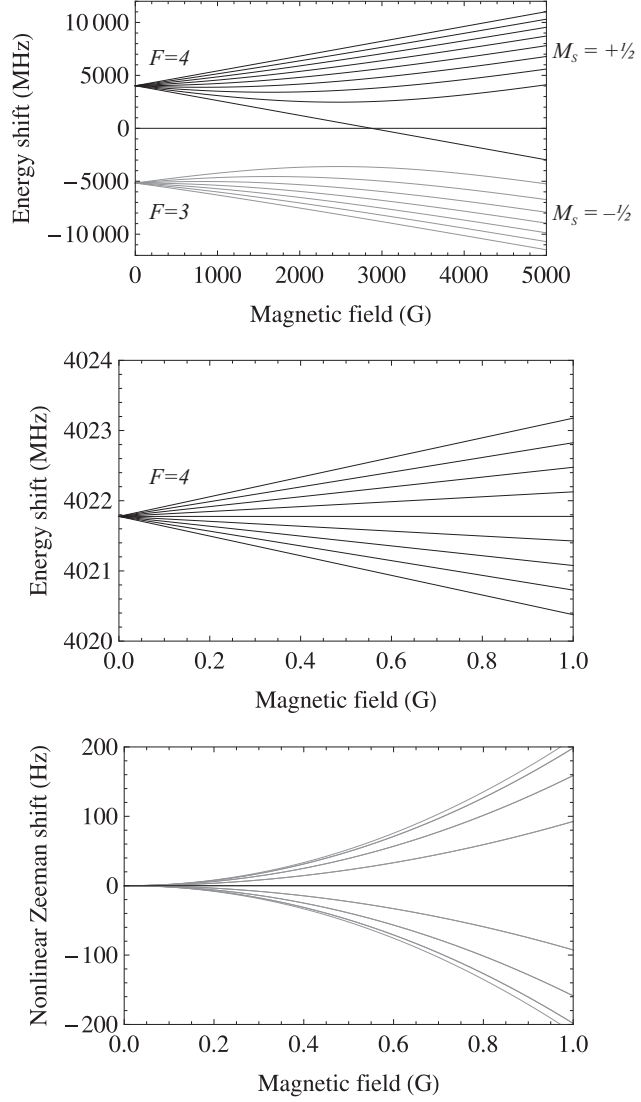
$$\boldsymbol{\mu} = g_F \mu_B \mathbf{F} , \quad (1.6)$$

where $\mathbf{F} = \mathbf{I} + \mathbf{J}$ is the total atomic angular momentum ($\mathbf{I}$ is the nuclear spin, and $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is the electronic angular momentum, $\mathbf{L}$ is the orbital angular momentum, and $\mathbf{S}$ is the electron spin) and g_F is the Landé factor corresponding to a state with a particular value of F . It is evident from symmetry that $\boldsymbol{\mu}$ is parallel to the total atomic angular momentum $\mathbf{F}$ (this follows, for example, from the Wigner-Eckart theorem – see the discussion in Refs. [28] and [29]). However, for larger magnitudes of $\mathbf{B}$ where the Zeeman shifts become significant compared to energy splittings caused by hyperfine and fine interactions, the magnetic moment $\boldsymbol{\mu}$ becomes dependent on the external magnetic field $\mathbf{B}$ due to mixing of atomic states (this is the nonlinear Zeeman effect, see Fig. 1.1 and the detailed discussion in Ref. [29]). Ultimately, then, the energy splitting ΔE between Zeeman sublevels in an atom, and consequently, the frequency Ω associated with time-evolution of the atomic angular momentum, is described by an expansion in powers of B :

$$\Delta E = \hbar\Omega = \gamma_0 + \gamma_1 B + \gamma_2 B^2 + \gamma_3 B^3 + \cdots = \sum_{n=0}^{\infty} \gamma_n B^n , \quad (1.7)$$

where the expansion coefficients γ_n depend on the particular atomic states and Zeeman sublevels involved. As seen in the lower plot of Fig. 1.1, for alkali atoms subjected to magnetic fields in the geophysical range of ~ 0.5 G, there are nonlinearities of the Zeeman shifts with magnitudes of tens of Hz, often larger than the magnetic resonance linewidths, that must be accounted for in sensitive optical magnetometry schemes.

Figure 1.1 Top and middle plots: energies of the ground-state $6s\ ^2S_{1/2}$ hyperfine manifold of cesium ($I = 7/2$) as a function of applied magnetic field (the Breit-Rabi diagram). At low fields (middle plot), the atomic states are well described by the coupled basis where F and M_F are good quantum numbers and the Hamiltonian (Eq. 1.4) representing the interaction with the applied magnetic field is treated as a perturbation. At high fields, the atomic states are well described by the uncoupled basis where M_I and $M_J = M_S$ are good quantum numbers and the hyperfine interaction is treated as a perturbation. Bottom plot: Nonlinear Zeeman shifts for the Cs ground state, computed by subtracting the linear term describing the low-field Zeeman shifts from the exact Breit-Rabi solutions.



1.1.3 Quantum beats and dynamic range

Since under all laboratory conditions Ω is very small compared to optical frequencies, and in many cases Ω is small compared to optical transition linewidths, Zeeman shift measurements typically involve observation of *quantum beats* [30–33]: the time-evolution of a coherent superposition of nondegenerate energy eigenstates at a frequency determined by the energy splittings. If a coherent superposition is created between two quantum states separated in energy by ΔE , according to the time-dependent Schrödinger equation, the complex phase between the states will

evolve according to $\varphi(t) = \Delta Et/\hbar$, or in the case of Zeeman shifts, as described by Eq. (1.7), $\varphi(t) = m\Omega t$ where m is an integer. The time-evolution of $\varphi(t)$ causes the complex index of refraction for light propagating through the atomic medium to evolve in time as well, which can be observed in the time-dependence of the intensity and polarization of the light field.

If all fields, magnetic and optical, applied to the atoms are constant or vary in time slowly compared to the ground-state spin-relaxation rate Γ_{rel} , the atomic spin polarization reaches a steady state. For spin-precession frequencies $\Omega \lesssim \Gamma_{\text{rel}}$, atomic spins are unable to undergo an entire quantum-beat cycle before relaxing. In this case, although spin precession angles accrued between pump and probe interactions are random, because the angles are small compared to π the polarization does not average to zero. This produces a residual, magnetic field dependent steady-state spin polarization that can be optically probed. For spin-precession frequencies $\Omega \gg \Gamma_{\text{rel}}$, transverse atomic spin polarization is averaged out because probed atomic spins have precessed by random angles greater than π with respect to one another. In other words, atoms polarized at different times will, in general, beat out of phase with each other, canceling out the overall atomic spin polarization. For this reason optical magnetometers with constant or slowly-varying fields have dynamic range limited by Γ_{rel} and find application as near-zero-field magnetometers.

Optical magnetometers used to measure fields corresponding to $\Omega \gg \Gamma_{\text{rel}}$ therefore generally employ some type of time-dependent field: magnetic field pulses to induce transient responses, radio-frequency (RF) fields to take advantage of magnetic-resonance techniques, or modulated light fields to synchronously optically pump the atomic medium. For example, synchronous optical pumping (or “optically driven spin precession” [34]) creates time-dependent macroscopic atomic spin polarization for high quantum-beat frequencies ($\Omega \gg \Gamma_{\text{rel}}$) by modulating the light at the quantum-beat frequency Ω or a subharmonic thereof. Polarization is produced in phase with that of atoms pumped on previous cycles, each optical pumping cycle contributing coherently to the atomic spin polarization, and as a result the ensemble beats in unison.

1.2 Model of an optical magnetometer

The basic principles of optical magnetometry are illustrated by considering an ensemble of two-level atoms with total angular momentum $F = 1/2$ in the ground state and $F' = 1/2$ in the upper state. Suppose that the atoms are immersed in a uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$ and left-circularly polarized (σ^+) light,² near-resonant with the (allowed electric dipole) $F \rightarrow F'$ transition, propagates along $\hat{\mathbf{x}}$

² Here we use the spectroscopists’ convention for left and right circular polarization, where a σ^+ photon (one with positive *helicity*, with the photon spin along its direction of propagation) is said to be left-circularly polarized, and a σ^- photon is said to be right-circularly polarized.

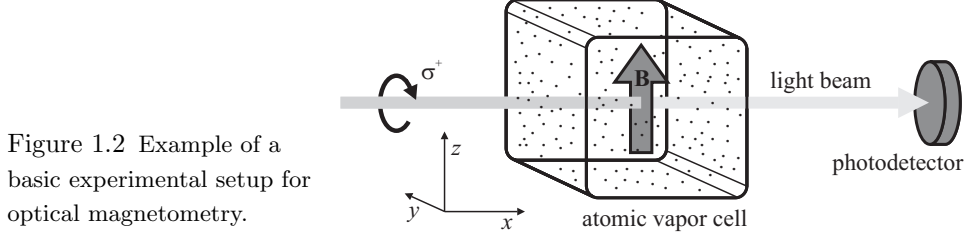


Figure 1.2 Example of a basic experimental setup for optical magnetometry.

(Fig. 1.2). The strength of the light-atom interaction is parameterized by the *Rabi frequency*

$$\hbar\Omega_R = d\mathcal{E}_0 , \quad (1.8)$$

where d is the transition dipole moment [for typical allowed electric-dipole transitions, $d \sim ea_0 \approx 2\pi \times 1.28 \text{ MHz}/(\text{V}/\text{cm})$] and $\mathcal{E}_0$ is the amplitude of the optical electric field. Spontaneous emission from the state F' back to F occurs at a rate Γ_0 (assume a closed transition). The optical pumping rate Γ_{pump} is given by Fermi's golden rule [28]:

$$\Gamma_{\text{pump}} = \frac{\Omega_R^2}{\Gamma_0} . \quad (1.9)$$

Let us assume that $\Gamma_{\text{rel}} \ll \Gamma_0$, $\Omega_R \ll \Gamma_0$, and ignore Doppler broadening (assume the atoms are stationary). Furthermore, to simplify our discussion, let us assume that $\Gamma_{\text{rel}} \ll \Gamma_{\text{pump}}$ (but note that for optimized magnetometer performance, typically $\Gamma_{\text{rel}} \gtrsim \Gamma_{\text{pump}}$). Under these conditions, the *saturation parameter* [28, 29]:

$$\mathcal{K} = \frac{\text{excitation rate}}{\text{relaxation rate}} = \frac{\Gamma_{\text{pump}}}{\Gamma_{\text{rel}}} = \frac{\Omega_R^2}{\Gamma_0 \Gamma_{\text{rel}}} \gg 1 , \quad (1.10)$$

indicates that the system is in the regime where the populations of ground-state Zeeman sublevels are strongly perturbed by the light field. Since $\Omega_R \ll \Gamma_0$, we can neglect stimulated emission and associated effects.

Suppose that initially $B = 0$ [Fig. 1.3(a)]. Interaction of the atoms with the light field causes ground-state optical pumping: choosing the quantization axis along the light propagation direction $\hat{\mathbf{x}}$, we see that the population of the $M_x = -1/2$ state (denoted $|-\rangle_x$) is depleted relative the population of the $M_x = +1/2$ state (denoted $|+\rangle_x$), since due to angular-momentum selection rules the light field only causes transitions between the $F = 1/2, M_x = -1/2$ and $F' = 1/2, M'_x = +1/2$ states. The atoms are optically pumped into a pure state $|+\rangle_x$ (since we can neglect ground-state spin relaxation under our assumptions) that does not interact with the light field. States that do not interact with light of particular polarization and/or frequency are commonly known as a *dark states*. The atomic medium has acquired net angular momentum along $\hat{\mathbf{x}}$ from the light field.

Suppose that after optically pumping the atomic ensemble into $|+\rangle_x$, the magnetic

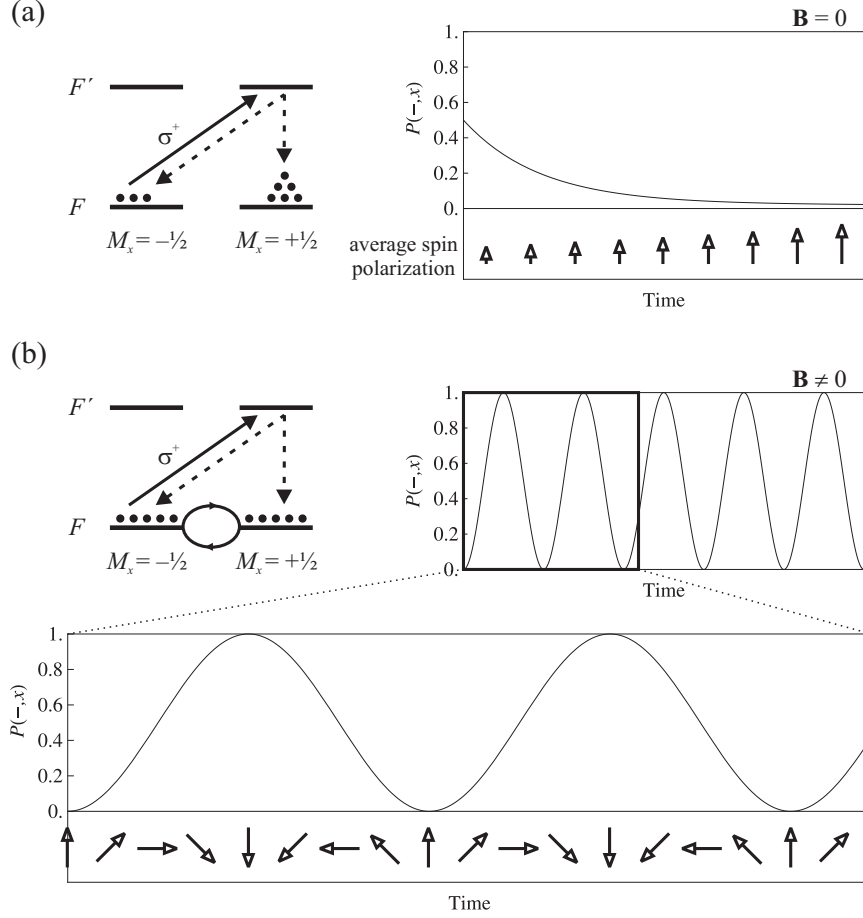


Figure 1.3 (a) Ground-state optical pumping of an ensemble of atoms with $F = 1/2$ in the ground state and $F' = 1/2$ in the upper state for the setup shown in Fig. 1.2 with $\mathbf{B} = 0$. Left-circularly polarized (σ^+) light excites the $M_x = -1/2 \rightarrow M'_x = +1/2$ transition, depleting the population of the $M_x = -1/2$ state. This decreases the probability $P(-, x)$ for atoms to be in the $M_x = -1/2$ state and increases the average spin polarization along x , as indicated by the increasing magnitude of the arrows along the bottom of the plot on the right. (b) Optical detection of spin precession with $\mathbf{B} \neq 0$. Larmor precession causes coherent oscillation between the $M_x = -1/2$ and $M_x = +1/2$ states (quantum beats) while the light probes the population of the $M_x = -1/2$ state [proportional to $P(-, x)$]. The coherent oscillation can also be understood as precession of the average spin polarization at the Larmor frequency, as shown by the arrows at the bottom of the lower plot.

field is suddenly (non-adiabatically) switched “on” ($B \neq 0$) and the light intensity is simultaneously reduced to a level sufficiently low so that the optical pumping rate is slow compared to the measurement time [Fig. 1.3(b)]. Consider the system from the point of view of a quantization axis chosen along $\hat{\mathbf{z}}$, parallel to $\mathbf{B}$: the energy of

the $M_z = +1/2$ state (denoted $|+\rangle_z$) is shifted by $+\hbar\Omega_L/2$ and the energy of the $M_z = -1/2$ state (denoted $|-\rangle_z$) is shifted by $-\hbar\Omega_L/2$. Using quantum mechanical rotation matrices [28, 29], it can be shown that

$$|+\rangle_x = \frac{|-\rangle_z + |+\rangle_z}{\sqrt{2}}, \quad (1.11)$$

$$|-\rangle_x = \frac{|-\rangle_z - |+\rangle_z}{\sqrt{2}}. \quad (1.12)$$

When the magnetic field is switched on at $t = 0$, the time-dependent state of the atomic ensemble (ignoring for now the light-atom interaction) is described by:

$$|\psi(t)\rangle = \sqrt{\frac{1}{2}} \left(e^{i\Omega_L t/2} |-\rangle_z + e^{-i\Omega_L t/2} |+\rangle_z \right). \quad (1.13)$$

The time-dependent probability $P(-, x)$ for an atom to be found in $|-\rangle_x$, a *bright state* which interacts strongly with the light field, is given by

$$P(-, x) = |\langle\psi(t)|-\rangle_x|^2 = \frac{1}{2} [1 - \cos(\Omega_L t)] = \sin^2(\Omega_L t/2). \quad (1.14)$$

This in turn modulates the light-atom-interaction probability and index of refraction at Ω_L , enabling optical detection of the quantum beats. The observed quantum beats are nothing but Larmor precession of the spin-1/2 atoms in the magnetic field cast in another language. The optical signal contains the information about the magnetic field B .

In the above model of an optical magnetometer, optical pumping occurs in a zero-field environment and optical probing occurs in the presence of the magnetic field to be measured. Practically, it is difficult and inefficient to null the magnetic field during optical pumping, and it is also frequently desirable to operate magnetometers continuously without temporal separation of pumping and probing stages.

An example of a magnetometry scheme enabling continuous optical pumping and probing of the atomic media in the presence of the magnetic field to be measured is the Bell-Bloom method [34]. Consider the same system as in the above model, however, in this case, $\mathbf{B}$ remains fixed throughout the measurement and the intensity of the light beam is sinusoidally modulated at a frequency Ω_m . Again, to simplify our discussion we assume that Γ_{pump} is fast compared to Ω_m and Γ_{rel} . At times when the light intensity is high, atoms are optically pumped out of $|-\rangle_x$ and into $|+\rangle_x$. The polarized atoms precess about $\hat{\mathbf{z}}$ at Ω_L , which is equivalently understood as coherent oscillation between $|+\rangle_x$ and $|-\rangle_x$ at Ω_L .

If the light intensity is modulated at $\Omega_m \neq \Omega_L$ (with $|\Omega_m - \Omega_L| \gtrsim \Gamma_{\text{rel}}$, which includes the case of unmodulated light where $\Omega_m = 0$), the precessing atomic polarization produced by atoms optically pumped from $|-\rangle_x \rightarrow |+\rangle_x$ at a particular time is not in-phase with respect to precessing atomic polarization produced at earlier times. However, when $\Omega_m = \Omega_L$, optical pumping from $|-\rangle_x \rightarrow |+\rangle_x$ happens synchronously with Larmor precession and precessing atomic polarization produced by

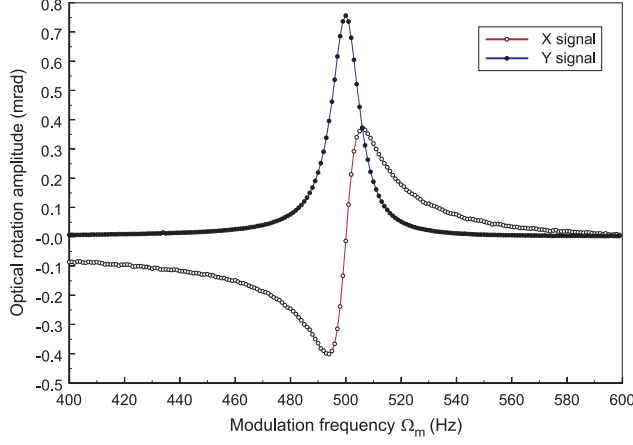


Figure 1.4 Characteristic resonant optical magnetometer signal used to measure the Larmor precession frequency. The X signal represents the signal amplitude in-phase with the frequency modulation and the Y signal represents the signal amplitude out-of-phase with the frequency modulation. Figure from Ref. [35].

atoms optically pumped at time t is reinforced by the atomic polarization optically pumped at $t - 2\pi/\Omega_L$, $t - 4\pi/\Omega_L$, $t - 6\pi/\Omega_L$, etc. Alternatively, one can say that in the “stroboscopic” picture where the atomic polarization is sampled only when the light intensity is at its peak the atomic spins are always oriented along $\hat{\mathbf{x}}$. Chapter 6 explores such optical-magnetometry techniques that employ modulated light, and similar techniques are also employed in helium magnetometers as discussed in Chapter 10.

Figure 1.4 shows a characteristic lineshape of the amplitude of an optical-magnetometer signal as Ω_m is tuned through resonance. In this particular example, linear polarization instead of circular polarization is used and $\mathbf{B}$ is along instead of perpendicular to the light propagation direction, but the essential physics remains similar (see Refs. [21, 33] and Chapter 6 for more details). For these data (from Ref. [35]), a laser beam tuned to the high-frequency side of the Doppler-broadened $F = 3 \rightarrow F'$ component of the D2 transition in atomic ^{85}Rb propagates through an antirelaxation-coated, buffer-gas-free Rb vapor cell. The laser is frequency modulated at Ω_m with a modulation amplitude of 65 MHz, which in turn modulates the light-atom interaction probability at Ω_m . A polarimeter measures the optical rotation of the laser light transmitted through the cell, and the oscillating signal is demodulated at the first harmonic of Ω_m with a lock-in amplifier. The characteristic resonant lineshape can be understood in analogy with a driven, damped harmonic oscillator: the atomic spin polarization produced by optical pumping naturally precesses at Ω_L due to the presence of $\mathbf{B}$, and when optical pumping (the driving term for the oscillator) is modulated at $2\Omega_L$ (the factor of 2 arises because of the two-fold symmetry of atomic spin polarization generated by linearly polarized light (see Refs. [29, 44, 45] and Sec. 1.3), there is a resonant enhancement of the atomic polarization leading to an enhancement of the optical-rotation signal. As is the case

with driven, damped harmonic oscillators, the phase of the time-dependent optical rotation depends on the detuning of the drive frequency from the natural oscillation frequency ($2\Omega_L - \Omega_m$), thus signals are observed both in-phase (X signal) and out-of-phase (Y signal) with the modulation of the light-atom interaction probability. While the X signal nominally crosses zero at $2\Omega_L = \Omega_m$, the Y signal is maximum when $2\Omega_L = \Omega_m$. This is because when the optical-pumping rate is synchronized with the spin-precession rate, the axis of the atomic spin polarization is parallel with the light polarization at the periodic maxima in the modulated light-atom interaction probability — when the atomic spin polarization axis is parallel with the light polarization no optical rotation is produced. The maximum optical rotation occurs when the atomic alignment axis is rotated by an angle $\phi = \pi/4$ with respect to the light polarization, which on resonance ($2\Omega_L = \Omega_m$) causes optical rotation out-of-phase with the modulation of the light-atom interaction probability.

For operation in varying ambient magnetic fields, optical magnetometers require a feedback loop to keep Ω_m locked to resonance. One approach is to use a phase-sensitive detector with an external feedback loop and a voltage-controlled oscillator. Another approach, often technically simpler, is to operate the magnetometer in a *self-oscillating* mode where the spin-precession signal is used directly to generate the oscillating RF-field (see Ref. [36] and Chapter 4) or modulation of the light (see Refs. [37–39] and Chapter 6).

A brief history of the development of optical magnetometers, ranging from the work of Michael Faraday in the 19th century to the recent developments described in this book, is presented in the online supporting material (www.cambridge.org/9781107010352).

1.3 Density matrix and atomic polarization moments

At the heart of all optical magnetometers are polarized atomic samples, whose properties can be modeled and understood using the *density-matrix* formalism. The density matrix makes it possible to describe ensembles³ of quantum systems that are more general than the ensembles which can be described by a single-atom wave functions (referred to as *pure* ensembles). The density matrix is defined in the following way. Consider the i^{th} atom in an ensemble described by the quantum mechanical state $|\psi_i\rangle$, which can be represented in terms of a set of orthonormal basis states $|m\rangle$:

$$|\psi_i\rangle = \sum_m |m\rangle \langle m|\psi_i\rangle . \quad (1.15)$$

³ An ensemble can either be assembled spatially (e.g., atoms contained in a vapor cell) or consist of sequential measurements separated temporally (under proper circumstances, this could even be a single quantum system).

The elements of the density matrix are given by

$$\rho_{m,n} = \sum_i \langle m | \psi_i \rangle \langle \psi_i | n \rangle . \quad (1.16)$$

The density-matrix formalism is discussed in detail in Refs. [29, 40].

The density matrix of an ensemble of atoms in a state with angular momentum F has $(2F+1) \times (2F+1)$ components $\rho_{M,M'}$, where M, M' refer to Zeeman sublevels. The density matrix ρ can be thought of as a tensor, and it is often useful to work with the irreducible components of ρ , i.e., the components $\rho_q^{(\kappa)}$ with $\kappa = 0, \dots, 2F$ and $q = -\kappa, \dots, \kappa$, which transform among themselves under rotations (see, for example, Refs [41] and [42]). One can represent ρ in the following way

$$\rho = \sum_{\kappa=0}^{2F} \sum_{q=-\kappa}^{\kappa} \rho_q^{(\kappa)} T_q^{(\kappa)}, \quad (1.17)$$

where $T_q^{(\kappa)}$ are irreducible tensor components represented as $(2F+1) \times (2F+1)$ matrices, and the coefficients $\rho_q^{(\kappa)}$ with $\kappa = 0, \dots, 2F$ and $q = -\kappa, \dots, \kappa$ are called *state multipoles* or *polarization moments* (PMs). The $\rho_q^{(\kappa)}$ are related to the $\rho_{M,M'}$ by

$$\rho_q^{(\kappa)} = \sum_{M,M'=-F}^F (-1)^{F-M'} \langle F, M, F, -M' | \kappa, q \rangle \rho_{M,M'} . \quad (1.18)$$

The following terminology is used for the different multipoles: $\rho^{(0)}$ – monopole moment (which is equal to the population divided by $\sqrt{2F+1}$), $\rho^{(1)}$ – vector moment or *orientation*, $\rho^{(2)}$ – quadrupole moment or *alignment*, $\rho^{(3)}$ – octupole moment, $\rho^{(4)}$ – hexadecapole moment, etc.⁴ Each of the moments $\rho^{(\kappa)}$ has $2\kappa+1$ components. Polarization moments can be visualized by plotting a three-dimensional surface where the distance of the surface from the origin in a given direction is equal to the probability of finding the highest projection of the angular momentum ($M = F$) in that direction [44, 45, 29].

The term *polarization* is used for the general case of an ensemble that has any moment with $\kappa > 0$. When the Zeeman sublevels are not equally populated, $\rho_0^{(\kappa)} \neq 0$ for some $\kappa > 0$, and the medium is said to have *longitudinal polarization*: polarization along the quantization axis. When there are coherences between the sublevels, $\rho_q^{(\kappa)} \neq 0$ for some $q \neq 0$, and the medium is said to have *transverse polarization*. When the magnetic field is along the quantization axis, transverse polarization precesses about $\mathbf{B}$, while longitudinal polarization remains static. For a given quantization axis z , the longitudinal orientation $\mathcal{O}_z$ and longitudinal alignment $\mathcal{A}_{zz}$ are

⁴ There are other definitions of the terms “orientation” and “alignment” in the literature. For example, in Ref. [43], alignment designates even moments in atomic polarization (quadrupole, hexadecapole, etc.), while orientation designates the odd moments (dipole, octupole, etc.).

given by

$$\begin{aligned}\mathcal{O}_z &\propto \rho_0^{(1)} \propto \langle F_z \rangle, \\ \mathcal{A}_{zz} &\propto \rho_0^{(2)} \propto \langle 3F_z^2 - F^2 \rangle,\end{aligned}\tag{1.19}$$

respectively.

Of particular interest for magnetometry [46–49] are the PMs with the highest possible κ for given electronic angular momenta L and S and nuclear spin I , which exhibit no nonlinear Zeeman effect. This is due to the fact that in the case of transverse polarization, PMs with maximal κ are related to coherences between stretched states, i.e., those states with maximal projections of angular momenta along the quantization axis, which are the same in the low- and high-field regimes (see Fig. 1.1 and surrounding discussion).

Also of importance is the role of spin-exchange collisions in the transfer of PMs between atoms: angular-momentum conservation demands that spin-exchange collisions preserve the net orientation in an atomic vapor, whereas higher-order PMs are not conserved in such collisions. Thus spin-exchange collisions inevitably lead to relaxation of PMs with $\kappa \geq 2$, as opposed to oriented atomic vapors where there are techniques to avoid spin-exchange broadening of magnetic resonances [50, 51].

Note that optical pumping with circularly polarized light (in the absence of other external fields) generally creates multipoles of all orders ($\kappa \leq 2F$), while pumping with linearly polarized light creates only even-ordered multipoles. This latter fact is a consequence of the symmetry of linearly polarized light: the optical electric field has no preferred direction in space, only a preferred axis. Also note that electric-dipole transition selection rules based on angular momentum conservation require that a single photon, having spin 1, can only possess PMs with $\kappa = 1$ (orientation) or $\kappa = 2$ (alignment). Thus to either generate or probe PMs with $\kappa > 2$ through light-atom interactions requires multi-photon processes, higher electromagnetic multipole transitions (for example electric quadrupole, etc.), or the presence of some other polarizing effect (such as thermal spin polarization in strong magnetic fields).

The optical properties of a polarized atomic medium are related to the symmetry of the PMs. A light field with wavevector $\mathbf{k}$ propagating in the $\hat{\mathbf{z}}$ direction can, in general, be written as a superposition of *optical polarization eigenmodes* (i.e., waves that traverse the medium without changing their state of polarization, experiencing only attenuation and phase shifts). The optical eigenmodes are determined by the symmetry of the medium, which in a spin-polarized vapor is given by the symmetry of the PMs. For example, in an oriented atomic vapor with light propagating along the orientation direction, the optical eigenmodes are left- and right-circularly polarized waves, since there is no other preferred axis or direction in the system. In fact, orientation, which is an axial vector,⁵ creates a preferred “handedness” in the system which manifests as a breaking of the symmetry between the two optical

⁵ An *axial vector* is a vector that does not change sign under spatial inversion.

eigenmodes. This symmetry breaking creates different complex indices of refraction for the two circular components.

Optical magnetometers often use measurements of optical rotation of linearly polarized light to monitor atomic spin precession. Linearly polarized light can be expressed as a coherent superposition of circular components, for example,

$$\hat{\mathbf{x}} = -\frac{1}{\sqrt{2}}(\hat{\mathbf{e}}_+ - \hat{\mathbf{e}}_-) , \quad (1.20)$$

$$\hat{\mathbf{y}} = \frac{i}{\sqrt{2}}(\hat{\mathbf{e}}_+ + \hat{\mathbf{e}}_-) , \quad (1.21)$$

where $\hat{\mathbf{e}}_+$ is the unit vector describing left-circular polarization and $\hat{\mathbf{e}}_-$ is the vector describing right-circular polarization (detailed discussion of the parametrization of light polarization is presented, for example, in Refs. [28, 29, 43]). The plane of light polarization is defined by the relative phase between the two circular components. As the light propagates through an oriented atomic vapor, because of the difference in the indices of refraction, the two circular components acquire a relative phase shift and the plane of light polarization is rotated. In this case, Larmor precession of the orientation generates oscillation of the optical rotation at the Larmor frequency. Similar effects occur when polarized light interacts with polarized atoms possessing other higher-rank PMs, such as alignment (see Chapter 6).

1.4 Sensitivity and accuracy

1.4.1 Variational sensitivity (short-term resolution) and long-term stability

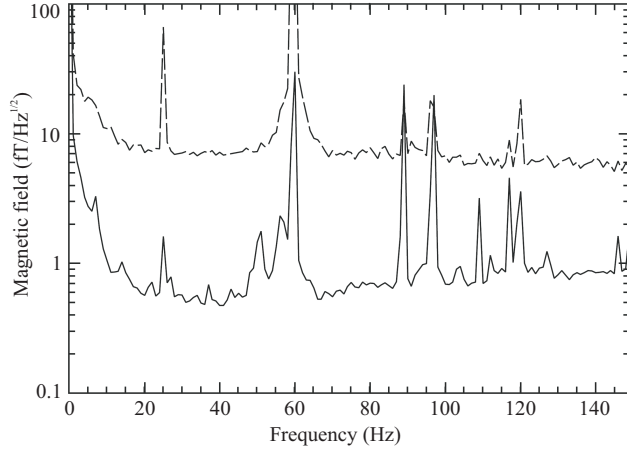
The (variational) sensitivity of a magnetometer in a stable field $\mathbf{B}$ is characterized by the dispersion δB of measurements averaged over a time interval T for a given duration τ of a single measurement (where $T \gg \tau$):

$$\delta B = \sqrt{\langle B(t)^2 \rangle - \langle B(t) \rangle^2} , \quad (1.22)$$

where $\langle \dots \rangle$ denotes the average of the measurements over time T .

Figure 1.5 shows an example of the spectrum of magnetic-field noise δB measured by a spin-exchange-relaxation-free (SERF) magnetometer using K atoms [51] (see Chapter 5). The magnetic-field-noise spectrum is obtained by recording the magnetometer response for ~ 100 s, performing a fast Fourier transform (FFT) without windowing, and calculating rms amplitudes in 1 Hz bins. The SERF magnetometer is a zero- or low-field optical magnetometer, and in this case, it is surrounded by a five-layer mu-metal magnetic shield system (see Chapter 12 for a review of magnetic-shielding techniques) and a system of coils null the residual fields and gradients. Residual magnetic field noise at the level of $7 \text{ fT}/\sqrt{\text{Hz}}$ is observed originating from Johnson noise of the innermost magnetic-shield layer. The intrinsic

Figure 1.5 Magnetic field noise spectrum measured by a single SERF magnetometer channel (dashed line) and intrinsic sensitivity (solid line) obtained by operating the SERF magnetometer in gradiometer mode (taking the difference between adjacent channels). Figure adapted from Ref. [51].



sensitivity of the magnetometer is obtained by canceling the ambient magnetic field noise through operation of the SERF magnetometer in gradient mode, where magnetic-field measurements carried out in two adjacent channels are subtracted (diffusion of K atoms is slowed by He buffer gas, enabling localized magnetic-field measurements — for the data in Fig. 1.5 the two channels are spaced 3 mm apart, so magnetic-field fluctuations are common-mode for the two channels).

Fundamentally, δB is limited by quantum fluctuations as noted in Sec. 1.1. However, in practical applications, technical sources of random noise contribute differently for different measurement durations τ , and thus, as in the study of atomic clocks, it is useful to study δB as function of the measurement duration τ . The time-dependence of sensitivity is usually described in terms of the Allan diagram [52] depicting the dependence of the Allan deviation $\sigma(\tau)$ or the Allan variance $\sigma^2(\tau)$ over the time interval $T = N\tau$:

$$\sigma^2(\tau) = \frac{1}{2(N-1)} \sum_{i=1}^N [B_{i+1}(\tau) - B_i(\tau)]^2, \quad (1.23)$$

where $B_i(\tau)$ represents the results of successive measurements within adjacent time intervals averaged over the duration τ . If the measurement results have a normal distribution, then the Allan deviation $\sigma(\tau)$ coincides with the sensitivity δB as expressed in Eq. (1.22).

Figure 1.6 shows the Allan deviation plot for a self-oscillating Rb magnetometer based on the double radio-optical resonance (DRR) technique [53] (see Chapter 4). The two stable isotopes of Rb, ^{85}Rb and ^{87}Rb , are contained within a single, antirelaxation-coated cell, and the Larmor precession frequencies f_{85} and f_{87} are simultaneously measured for both isotopes. The self-oscillating DRR Rb optical magnetometer is a high-field magnetometer, in the present example the ambient field is $B = 1170 \text{ nT} = 11.7 \text{ mG}$, stabilized with a feedback loop using the magne-

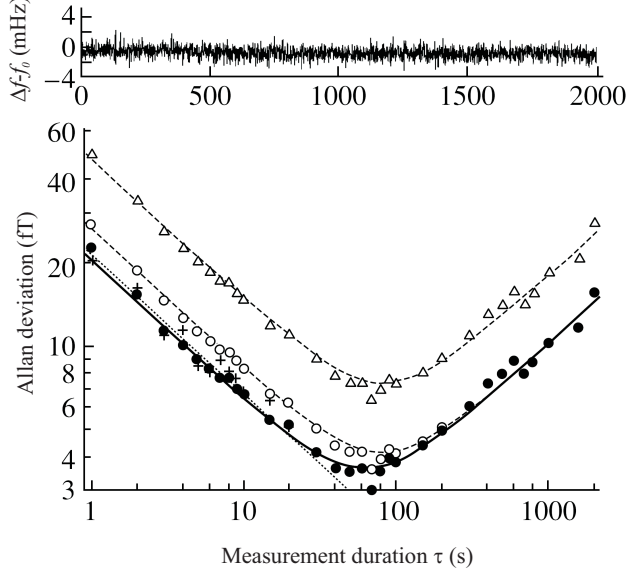


Figure 1.6 Upper plot: record of the difference frequency $\Delta f - f_0$ (see text). Lower plot: Allan deviation of the ^{87}Rb magnetometer signal (open triangles), the magnetic field derived from the difference frequency Δf (open circles), the ^{87}Rb magnetometer signal after subtraction of ambient field noise (filled circles); shot-noise-projected sensitivity (crosses). Figure adapted from Ref. [53].

tometric signal from ^{85}Rb . Based on the gyromagnetic ratios for the Rb isotopes, $3f_{85} \approx 2f_{87}$, and so the difference frequency $\Delta f = 3f_{85} - 2f_{87}$ is near zero with only a small offset $f_0 \approx 12.47$ Hz. This allows straightforward subtraction of the noise due to fluctuations of the ambient magnetic field yielding the intrinsic sensitivity of the optical magnetometer, which agrees well with the projected shot-noise-limited sensitivity for integration times less than 100 s.

A comparison of Allan diagrams of different magnetometric devices indicates that they are characterized (similarly to atomic frequency standards) by the measurement time τ_0 at which the decrease in the Allan deviation as a function of τ is followed by a flat dependence or an increase in $\sigma(\tau)$ as a function of τ (in Fig. 1.6, $\tau_0 \approx 80$ s). Such behavior is due to parametric shifts of the magnetic resonance frequency Ω itself and to errors in the measurement of its value. The variation in $\sigma(\tau)$ over measurement durations $\tau > \tau_0$ determines the long-term stability of the magnetometer.

1.4.2 Parameter optimization

As seen in the preceding examples (Figs. 1.5 and 1.6), the sensitivity of a magnetometer can be determined via direct measurement in a magnetic-field environment with magnetic-field stability exceeding the magnetometric sensitivity, or via a differential measurement using identical magnetometers measuring the field in approximately the same region of space. For parameter optimization, however, it may be more efficient to determine the response of the optical signal to a known

change in the magnetic field and estimate the sensitivity based on the light power reaching the detector. This approach separates the parameter optimization problem from the potentially difficult issues connected with eliminating technical sources of noise limiting the practical magnetometric sensitivity.

In parameter optimization studies, what is typically reported is the *shot-noise-projected* (SNP) sensitivity, given by:

$$\delta B_{\text{SNP}} = \left(\frac{d\varphi}{dB} \right)^{-1} \delta\varphi, \quad (1.24)$$

where $d\varphi/dB$ is the slope of the detected optical signal with respect to the applied magnetic field and $\delta\varphi$ is the photon-shot-noise projected sensitivity of the optical measurement. It should be noted that in most cases δB_{SNP} cannot be reduced below the atomic-shot-noise limit given by Eq. (1.1) (important exceptions are described in Chapters 2 and 3). As seen in Fig. 1.6, $\delta B_{\text{SNP}} = \delta B$ when ambient magnetic-field noise can be reduced below the sensitivity of the magnetometer.

1.4.3 Absolute accuracy and systematic errors

One of the key advantages of optical magnetometers as compared to other magnetometry schemes is that, in principle, measurement of the spin-precession frequency Ω yields an intrinsically accurate, calibration-free determination of the magnetic field. Nonetheless, there are several sources of systematic errors that need to be considered in magnetometric measurements (see also Chapter 20 for further discussion of systematic effects).

One important source of systematic error is the effect of the pump and probe light on the precession frequency. The light-atom interaction can shift the energy of Zeeman sublevels through the AC Stark effect, and hence shift Ω . This produces, in effect, a fictitious magnetic along the axis of light propagation proportional to the degree of circular polarization of the light [54]. There are several techniques that can be used to reduce light shifts: (1) reduce the light intensity, since the AC Stark shifts generally scale proportionally for sufficiently low intensities; (2) employ an experimental geometry where the light propagation direction is orthogonal to the measured magnetic field direction, in which case the fictitious field adds in quadrature with the leading field; (3) employ linearly polarized light, which in principle eliminates the effect, although there can be issues with birefringence of optical elements in between the polarizers and the atomic sample, as well as subtle issues related to alignment-to-orientation conversion (AOC) that can produce fictitious fields even from linearly polarized light [55, 35]; and (4) tune the laser light to a frequency where the AC Stark shifts are minimized, taking advantage of zero-crossings in the spectral dependence of the AC Stark effect [56, 57].

While the atomic spins are precessing in the magnetic field they are effectively decoupled from the experimental apparatus, and thus rotations of the apparatus

can lead to gyroscopic shifts of the precession frequency (Chapter 19). This is especially important for optical magnetometers on moving platforms. Even gyroscopic effects due to the rotation of the earth can be significant for stationary optical magnetometers used in high-precision tests of fundamental physics [58, 59] (see Chapter 18).

Because of the vector/tensor nature of the light-atom interaction employed for pumping and probing atomic spins, there are typically certain orientations of the magnetic field relative to the magnetic sensor (in particular relative to the light propagation direction and light polarization axis/direction) for which the magnetometric sensitivity is drastically reduced. Such orientations are known as *dead zones*, and the problem has been addressed by using multiple sensors [60, 61], mechanical rotation of some components [62], use of unpolarized light and spatially varying microwave fields [63], and more recently by simultaneously pumping and probing PMs of different κ [64]. Furthermore, due to shifts, splittings, and lineshape asymmetries of magnetic resonances caused by the nonlinear Zeeman effect, the measured magnetic field magnitude can acquire a false dependence on the relative orientation of the sensor to the field (*heading error*).

Collisions between atoms and with cell walls also can cause systematic shifts of Ω . Collisional frequency shifts are well-studied [1–3] and can be accounted for with precise knowledge of the atomic vapor density, while frequency shifts due to wall collisions can be minimized with a spherical cell geometry.

1.5 Vector and scalar magnetometers

Optical magnetometers can be operated in either *scalar mode*, measuring the magnitude of the magnetic field, or *vector mode*, measuring one or more vector components of the field. There are advantages and disadvantages for each mode of operation, and so the choice of configuration largely depends on the application. Scalar magnetometers are insensitive to rotation of the magnetometer in the Earth’s magnetic field and rely on frequency measurements which can be made with very high accuracy. For example, a 10 nG change in the Earth’s magnetic field corresponds to a fractional frequency shift of 20 ppb, easily within the stability range of even the simplest quartz oscillators. Since the relation between the frequency and the magnetic field [see Eqs. (1.5) and (1.7)] depends only on fundamental constants, scalar atomic magnetometers do not require calibration and are essentially free from aging effects.

Vector magnetometers can measure all three components of the magnetic field and thus can obtain more complete information about the field and provide much better localization information for detection of magnetic-anomaly sources [65]. On the other hand, vector magnetometers are much more sensitive to rotations: in the worst case, a rotation by ~ 20 nrad will mimic a field change of ~ 10 nG. Therefore,

vector magnetometers have to be mounted on stationary platforms, combined with sensitive gyroscopes, or used in combination to form magnetic gradiometers in order to realize magnetometric sensitivities near the $\text{nG}/\sqrt{\text{Hz}}$ level in practical systems.

Optical magnetometers can naturally operate in either vector or scalar mode using essentially the same setup [66–68]. The technique for conversion between scalar and vector operation relies on the fact that if a small bias field is applied to the sensor in a certain direction in addition to the field to be measured, then the change in the overall field magnitude is linear in the projection of the bias field on the main field, and is only quadratic (and generally negligible) in the projection on the orthogonal plane. Thus, applying three orthogonal bias fields consecutively, and performing three measurements of the overall magnetic field magnitude, one reconstructs the overall field vector.

1.6 Applications

The applications of optical-magnetometry techniques are diverse and expanding. Optical magnetometers are used to carry out a variety of geophysical magnetic field measurements (discussed in Chapter 17) — studying natural and anthropogenic magnetic anomalies, investigating geomagnetic-field dynamics, and measuring the magnetic properties of geologic samples. Optical magnetometers are also deployed to measure magnetic fields in space (Chapter 15) and new techniques have recently been proposed to measure geomagnetic fields using the naturally occurring atomic-sodium rich layer in the mesosphere (Chapter 13). Of considerable interest are direct measurements of magnetic fields of biological origin (Chapter 16), for example those from the heart and brain. A variety of state-of-the-art optical magnetometers with extremely high spatial resolution, in particular those employing nitrogen-vacancy (NV) centers in diamond (Chapter 8) and spinor Bose-Einstein condensates (Chapter 9), are now capable of magnetic microscopy enabling new applications in material science and biology. Optical magnetometers have also been applied to novel nuclear magnetic resonance (NMR) and magnetic resonance imaging (MRI) studies (Chapter 14). Already significant progress has been made in miniaturization of optical magnetometers, to the point where microfabrication techniques allow chip-scale devices (Chapter 7). Related experimental techniques are also used in tests of fundamental physical laws and symmetries of nature (Chapter 18) and for gyroscopes based on atomic spin (Chapter 19). It is exciting to witness and participate in moving optical magnetometers from the laboratory domain to a variety of diverse applications.

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